# Module 13

## Comparing Remote Access Methods

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# Learning Objectives

- Summarize WAN provider and Internet access types.
- Compare and contrast VPN topologies and protocols.
- Explain remote host access and management methods.

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# Lesson 13.1

## WAN and Internet Connectivity

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# Wide Area Networks (WANs)

## WANs vs. LANs

- WANs: Data communication over large distances, often using public networks
- LANs: Cover shorter, localized areas

## Enterprise WAN

- Network for a single organization spanning large geographic areas
- Uses public telecom infrastructure for wide-reaching connectivity

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# WANs and the OSI Model

## WAN Physical Layer

- Defines mediums and interfaces for data transmission
- Uses modems for copper cable connections (analog and digital)

## Data Link Layer in WANs

- Simpler protocols compared to Ethernet in LANs
- Ethernet increasingly used for efficient end-to-end connectivity

## The Network Layer

- Uses Internet Protocol (IP) for addressing
- Connects customer and provider sites with CE and PE routers
- Assigns public IPv4 or IPv6 addresses for global identification

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# Internet Types – Accessibility and Speed

**T1 Leased Lines**

- High reliability; dedicated bandwidth
- Speed up to 1.544 Mbps

**DSL (Digital Subscriber Line)**

- Widely available; uses existing phone lines
- Speed varies; ADSL2+ up to about 24 Mbps downlink

**Cable Internet**

- Broad availability in urban areas
- Speed up to 38 Mbps downlink (DOCSIS standards may allow higher)

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# Internet Types – Technology and Usage

## T1 Leased Lines

- Uses time-division multiplexing
- Ideal for businesses needing constant connection

## DSL (Digital Subscriber Line)

- Offers both asymmetrical (ADSL) and symmetrical (SDSL) options
- Versatile for both residential and small business use

## Cable Internet

- Hybrid Fiber Coax (HFC) technology
- Suitable for heavy residential or business internet usage

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# Internet Types – Implementation and Cost

## T1 Leased Lines

- Requires professional installation; higher monthly cost
- Scalable with additional lines

## DSL (Digital Subscriber Line)

- Easy setup with existing phone line; cost-effective for home use
- Performance decreases with distance from provider

## Cable Internet

- Installation varies; generally lower cost than T1
- Shared bandwidth can reduce speed during peak times

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# Fiber to the Curb

## Fiber to the Curb (FTTC)

Fiber optic cables run directly to the curbs near homes/ business environment

## Benefits

High-speed internet access compared to traditional copper lines

Increased reliability and lower latency

## Considerations

Requires a final connection to the premise via coaxial cable or copper phone lines

Slightly lower speeds than FTTP due to the last leg being non-fiber

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# Fiber to the Premises

## Definition

Fiber optic connection is run directly into the premises

## Benefits

Highest possible speed of internet service by eliminating non-fiber connection points

Future-proof infrastructure with capabilities for expanding bandwidth needs

## Considerations

Higher installation costs compared to FTTC

May be unavailable in all areas due to the extensive infrastructure required

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# Activity: True or False

Customer and provider sites are linked via customer edge (CE) and provider edge (PE) routers, with the assignment of public IPv4/IPv6 addresses.

True or false?

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# Lesson 13.2

## Comparing Remote Access Methods

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# Remote Access Policy

## Remote Access Considerations

- Authorized access only, strong identity verification
- Use VPNs, encrypt data in transit
- Secure remote devices, regular updates, anti-malware, security policies

## Implementing a Remote Access Policy

- Role-based access, resource/time limits
- Detailed access logs, review for suspicious activities
- Regular security updates for RAS/network devices, secure management access

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# Tunneling Protocols Overview

## Tunneling Protocols

- Secure private data passage across public networks
- Encapsulate packets within packets for security
- Create secure tunnels for data
- Ensures sender-receiver exclusive access

## Layers Operation

- Varying OSI model layers
- Tailored deployments for security, speed, compatibility

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# Key Tunneling Protocols

| Point-to-Point Protocol (PPP) | GRE (Generic Routing Encapsulation) | IPSec (IP Security) | TLS & DTLS (Transport Layer Security & Datagram) | Application |
|---|---|---|---|---|
| Data Link layer operation | Network layer function | Network layer focus | Session layer operation | VPN use for encrypted, authenticated remote communication |
| Used for digital lines, lacks inherent security | Wide protocol encapsulation capability | Secures IP via authentication, encryption | Security with increased overhead | |

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# Internet Protocol Security (IPSec) Modes

## Transport Mode

- Used primarily within a private network
- Encrypts only the payload of the IP packet
- Authentication Header (AH) provides IP header integrity
- Encapsulating Security Payload (ESP) encrypts the payload for confidentiality

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# Internet Protocol Security (IPSec) Modes

## Tunnel Mode

- Ideal for VPNs and securing communications across public networks
- Encrypts the entire IP packet, including the header
- Entire packet treated as the payload and encapsulated within a new IP packet

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# Configuring IPSec

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# Internet Key Exchange (IKE)

## Internet Key Exchange (IKE)

- Protocol in IPSec for secure communication over the Internet
- Essential for secure, reliable VPN infrastructure
- Combines strengths of IKEv and IKEv2

## Purpose

- Negotiation and establishment of SAs
- Mutual authentication
- Exchange of cryptographic keys

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# Internet Key Exchange (IKE)

## IKEv1

- Site-to-site and host-to-host VPNs
- Secure, authenticated communication channel
- Supporting protocol for remote access VPNs
- Flexibility in key management and security policy agreements

## IKEv2

- Supports EAP for user/device authentication
- Simplified setup, better NAT traversal, MOBIKE for mobile connectivity
- Streamlines secure connection establishment
- Preferred for modern VPN solutions

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# Client-to-Site VPN Overview

## Definition

- VPN topology where individual clients connect to a corporate network over public Internet

## Components

- VPN Client: Software or device initiating the connection
- VPN Gateway: VPN-enabled router at the network's edge

## Use Case

- Ideal for homeworkers and field employees

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# VPN Protocols and Security

## Protocols Used

- SSL/TLS VPN: Secure tunneling with certificates (e.g., SSTP)
- L2TP/IPSec: Widely used for strong security

## Security Concerns

- Deprecated Protocol: PPTP, due to security flaws

## Traffic Management

- Split Tunnel vs. Full Tunnel: Balancing direct Internet access and security needs

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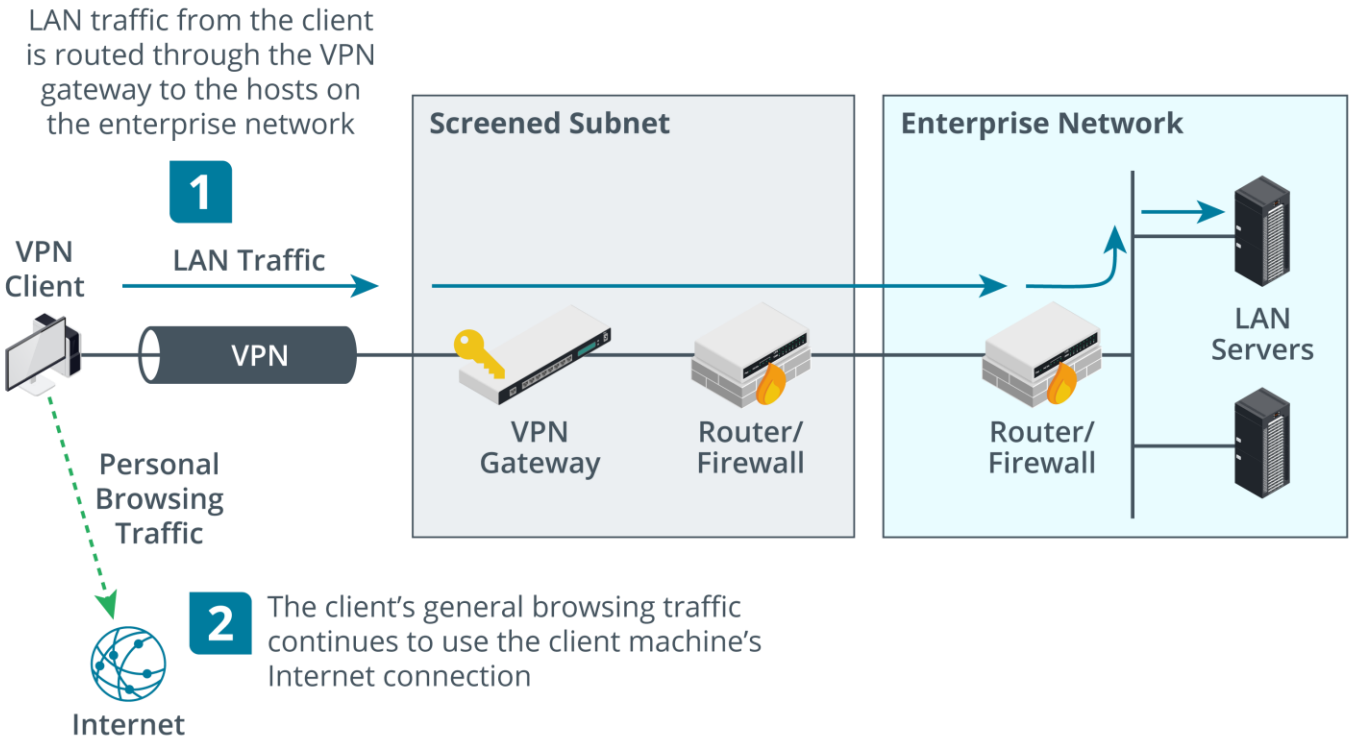
# Split Tunnel Example

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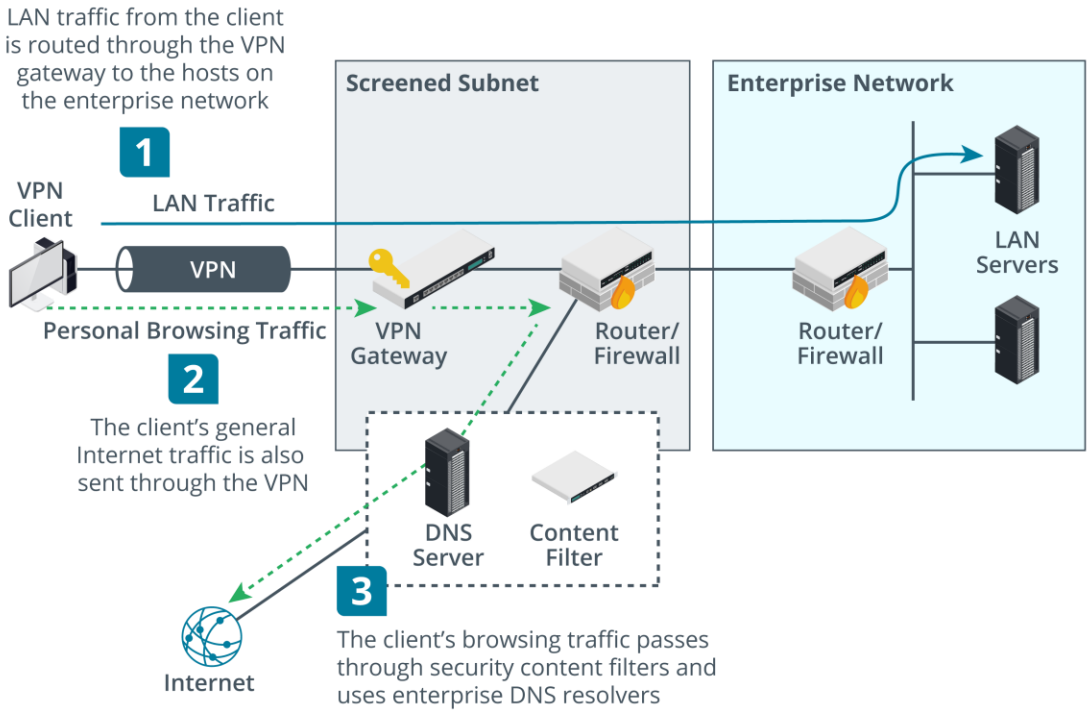
# Full Tunnel Example

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# Clientless VPN

## Definition

- Allows access to network resources via web browser
- Does not need dedicated VPN client software
- Uses HTML5's canvas element and WebSockets

## Benefits

- Simplifies access across devices
- Enhances security by isolating the session within the browser
- Offers a cost-effective solution

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# Clientless VPN Use Cases and Examples

## Use Cases

- Remote desktop access using Microsoft RDP or similar protocols
- Secure access to web applications and virtual desktops through a remote desktop gateway

## Examples

- Microsoft Remote Desktop Services and Citrix Products for enterprise solutions
- Apache Guacamole for open-source implementations

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# Site-to-Site VPN

| Automatic Connectivity | High Security | Simplified Management | Scalability |
| --- | --- | --- | --- |
| • Always on, providing seamless data flow between networks | • Advanced encryption ensures secure data sharing across networks | • Reduces the need for individual device configuration | • Scales to accommodate new sites or networks with minimal infrastructure changes |

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# Site-to-Site VPN Use Cases

| Cross-Office Connectivity | Mergers and Acquisitions | Industrial IoT (IIoT) Networks | Cloud Integration |
|---|---|---|---|
| • Securely connects & shares resources across locations | • Facilitates the integration of IT infrastructure between companies | • Protects corporate networks & IIoT data with secure links | • Securely connects a company's on-premises network to cloud services |

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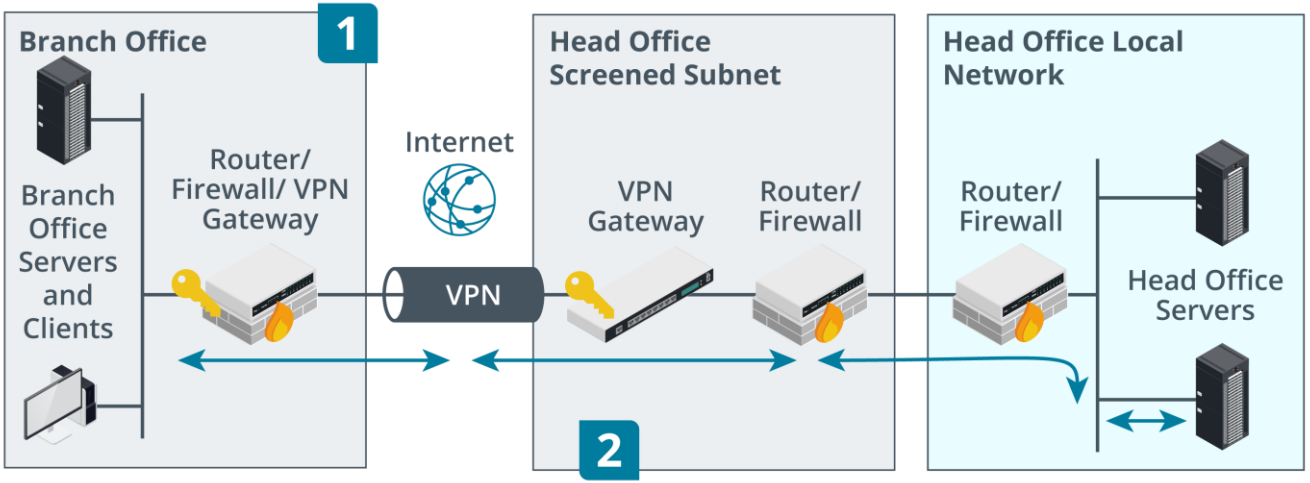
# Site-to-Site VPN Example

The VPN gateway at a branch office establishes a VPN connection with the head office site




Traffic for a host at a remote site is automatically routed and tunneled over the VPN link

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# Activity: Two Truths and a Lie

IKEv1 is used for site-to-site and host-to-host VPNs

Site-to-site VPNs are ideal for homeworkers and field employees

Clientless VPNs allow users to access network resources via a web browser

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# Lesson 13.3

## Remote Management

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# Remote Host Access

## Remote Host Access

- Enables remote control of computers and network devices

## Use Cases

- Configuring network appliances remotely
- Manage servers or users
- Operate computers from afar

## Technologies

- Secure Shell (SSH) for secure command-line access
- Remote desktop protocols for graphical interface access
- Remote desktop gateways for connecting to networked applications or virtual desktops, facilitating clientless VPNs

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# Secure Shell (SSH)

## Secure Shell (SSH)

- Protocol for secure communication between devices
- Primary use: remote access to servers and network equipment
- Enables secure file transfers, terminal sessions, and command execution
- Strong encryption (confidentiality and data integrity)

## Benefits of SSH

- All communications are encrypted
- Supports various authentication methods
- Compression of data during transfer speeds up remote management
- Available on all major operating systems

## Components

- SSH Server and Client
- Authentication methods
- SSH keys
- SSH commands

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# SSH Commands

**ssh**

Connects to a remote server to initiate a secure shell session

**ssh-keygen**

Generates a new SSH key pair (private and public keys) for user authentication

**ssh-copy-id**

Copies a public SSH key to a remote server's authorized keys file

**ssh-add**

Adds private key identities to the authentication agent

**scp**

Securely copies files between hosts on a network

**sftp**

Secure File Transfer Protocol, used for secure file transfer over SSH

**ssh-agent**

Holds private keys used for public key authentication and manages SSH keys

**sshd**

The SSH Daemon, which listens for incoming SSH connections on the server

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# Telnet

## Telnet

- Terminal emulation (protocols & shell commands): Remote system access

## Telnet Daemon

- A service running on the remote computer, listening by default on TCP port 23

## Security Concerns

- Communications via Telnet, including passwords, are not encrypted
- Vulnerable to packet sniffing and replay attacks
- Previously used for configuring network equipment (switches and routers)

## Best Practices

- Avoid using Telnet for secure tasks
- Disable or uninstall Telnet service and block port 23
- Deploy in secure network segment (for managing legacy systems)
- Consider SSH or other encrypted protocols for remote management tasks

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# Remote Desktop Protocol (RDP)

| RDP | Key Features | Security Considerations |
| --- | --- | --- |
| • Protocol designed by Microsoft<br>• Control desktops remotely via network connection | • Uses TCP port 3389<br>• Supports graphical user interface (GUI)<br>• Secured with encryption and Network Level Authentication (NLA) | • Vulnerable to malware if connecting to compromised servers<br>• RDP Restricted Admin (RDPRA) Mode and Remote Credential Guard for enhanced security |

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# Remote Desktop Protocol (RDP)

## Best Practices

- Avoid using RDP from low-trust workstations
- Use a secure administrative workstation (SAW) for remote connections
- Implement RDP for application virtualization and remote administration

## Alternatives to RDP

- TeamViewer
- Virtual Network Computing (VNC)
- Citrix's products for cross-platform remote access

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# Console Connections and OOB

## Console Connections

- Direct connection for network device configuration/ troubleshooting
- Console (or rollover) cable connected to a laptop with terminal emulator software
- Configuring devices when network access is unavailable or for recovery purposes

## Out-of-Band (OOB) Management

- Dedicated channel for secure device administration
- Secure and reliable access to device's administrative functions, especially during network outages

## Examples of OOB Interfaces

- AUX port for dial-up access
- Management port for IP-based management separate from data traffic

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# Jump Boxes

## Jump Box

- Secure AP to manage servers and devices in screened subnet/cloud network
- Acts as an entry point
- Reduces the potential attack surface

## Benefits

- Enhanced security
- Simplified management

## How It Works

- Admins connect to the jump box using secure protocols (SSH or RDP)
- From the jump box, access the admin interfaces on application servers
- Application servers configured to only accept connections from the jump box, enhancing security

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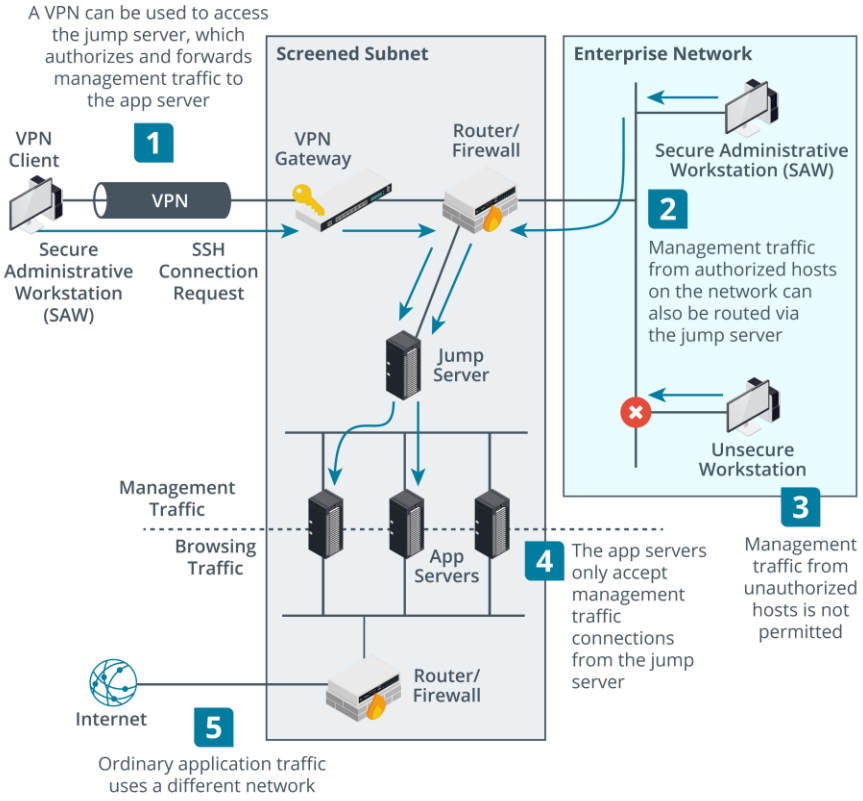
# Jump Boxes Example

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# API Connection Methods

## Automation & Orchestration

- Use scripts for configuring appliances
- Avoid manual setups

## APIs in Action

- Google's virtual firewall example
- Accessible via web console, CLI, or API
- Demonstrates deployment flexibility

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# API Connection Methods - Security

## Security Practices

- Enforce strong authentication policies
- Avoid root user for automation
- Use secret keys for programmatic access
- HTTPS for API communication

## Least Privilege Principle

- Automation accounts with minimum rights

## Secure Communication

- Importance of HTTPS
- Mutual authentication for API security

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# Activity: Name That Command

1. Connects to a remote server to initiate a secure shell session

2. Securely copies files between hosts on a network

3. Adds private key identities to the authentication agent

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# Summary

- Evaluate WAN options (leased line, DSL, cable, FTTx)
- Develop remote access policy for secure connections
- Enable VPNs/remote desktop with client compatibility & secure firewall
- Secure critical access with SSH/RDP over out-of-band & consider programmatic/API access

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